

Lecture Notes

Stochastic Hydrology

Module 1: Probability and Statistics

Prof. Dr. Paola Passalacqua

Chair of Hydrology and River System Science
Institute of Environmental Engineering
ETH Zürich

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Preface

The content and structure of these lecture notes is based on the following works:

Statistical Methods in Water Resources

Book 4, chapter A3. U.S. Geological Survey. 2002. Helsel, D.R. and R. M. Hirsch. DOI

Handbook of Hydrology

McGraw-Hill, Inc. 1992. Maidment, David R., ed.

These lecture notes are intended for personal use in the Stochastic Hydrology course and may not be distributed further.

1 Random Variable, Probability Distributions & Descriptive Statistics

Hydrologic processes evolve in space and time in a manner that is partly predictable, or deterministic, and partly random. This type of process is called a stochastic process. The main terms in statistics are:

- A *random variable* X is the outcome of an experiment, a process, etc. that has an element of uncertainty associated with it and is described by a probability distribution.
- The *probability distribution* specifies the chance that an observation x of the variable will fall in a specified range of X .
- A finite set of observations $x_1, \dots, x_n$ of the random variable is called a *sample*. It assumes that samples are drawn from a hypothetical infinite *population* possessing constant statistical properties, while the properties of a sample may vary from one to another.
- The set of all possible samples that could be drawn from the population is called the *sample space*, and an *event* is a subset of the sample space.
- The *probability* $\mathbb{P}(A)$ is a measure of the likelihood of occurrence of an event A when an observation of a random variable is made.

1.1 Frequency and Probability Functions

1.1.1 Probability Density Function

The probability density function PDF, denoted by $f_X(x)$, describes the distribution of probability associated with the values a random variable X can assume.

Properties of PDFs:

- *Non-negativity*: The probability density function is non-negative for all real values.

$$f_X(x) \geq 0 \quad \forall x \in \mathbb{R} \quad (1.1)$$

- *Normalization (Unit Total Probability)*: The total probability obtained by integrating the density over all possible outcomes is equal to one.

$$\int_{\mathbb{R}} f_X(x) dx = 1 \quad (1.2)$$

The probability density function is defined for a population. The corresponding function for a sample is the *relative frequency function*, which is visualized with a *histogram*. A visualization of both the relative frequency function and the probability density function for total annual precipitation measurements are shown in Figure 1.1a) and b), respectively.

Creating a histogram:

1. Divide the range of values into m classes $[x_1, x_2], (x_2, x_3], \dots, (x_{m-1}, x_m]$ of width Δx . The number of classes can be estimated by different methods, including Sturge's Rule, where $m = 1 + 3.3 \cdot \log_{10}(N)$ is based on the total number of observations N .
2. Count the number of data values falling into each class n_i .
3. The number of observations falling into a class n_i divided by the total number of observations N is the relative frequency

$$\frac{n_i}{N} = \frac{\# \text{ observations in class } i}{\text{total } \# \text{ of observations}} = \text{relative frequency for class } i \quad (1.3)$$

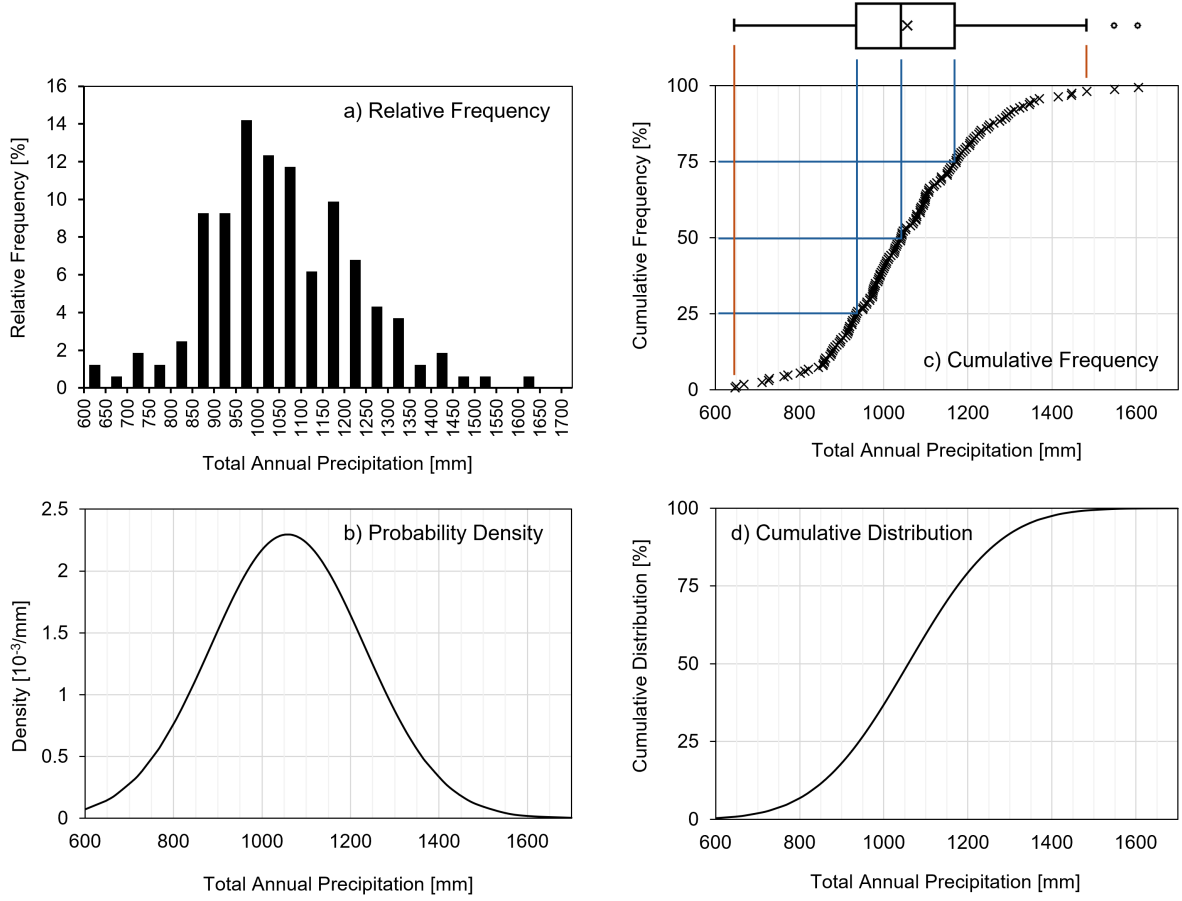


Figure 1.1. Visualization of the total annual precipitation from the Zürich Fluntern measuring station from 1865-2024, including the sample (top) and population (bottom) figures assuming a normal distribution, and relative (left) and cumulative (right) representations. On the top right, the box-and-whiskers representation and its connection to the cumulative frequency distribution is shown.

1.1.2 Cumulative Distribution Function CDF

The cumulative distribution function $F_X(x)$ represents the probability that a given random variable X is less than or equal to a specified value x and therefore describes the non-exceedance probability.

Properties of CDFs:

- Definition of the cumulative distribution function CDF: Non-exceedance probability

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{\infty} f_X(t) dt \quad (1.4)$$

- Exceedance Probability:

$$1 - F_X(x) = \mathbb{P}(X > x) \quad (1.5)$$

- Relationship between CDF and PDF:

$$dF_X(x) = f_X(x) dx \quad \leftrightarrow \quad f_X(x) = \frac{dF_X(x)}{dx} \quad (1.6)$$

- Probability over an interval:

$$\mathbb{P}(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} f_X(t) dt = F_X(x_2) - F_X(x_1) \quad (1.7)$$

The cumulative distribution function is defined for a population. The corresponding function for a sample is the *cumulative frequency function*. A visualization of both the cumulative frequency function and the cumulative distribution function for the total annual precipitation measurements is shown in Figures 1.1c) and d), respectively.

Creating a cumulative frequency plot:

1. Sort the data in ascending order.
2. Assign a rank to each observation $i = 1, 2, \dots, N$, where N is the total number of observations.
3. Plot the data against its empirical plotting position. The plotting position can be defined by different approaches, including the simple Weibull plotting position $F(i) = i/(N + 1)$.

The resulting cumulative frequency distribution shows the fraction of data with values smaller than a given threshold.

1.2 Moments of a Distribution and Descriptive Statistics

In stochastics, *moments* are quantitative measures that describe the shape and characteristics of a probability distribution of a variable. The r th moment about the origin (also called the raw moment) is defined as $\mathbb{E}[X^r]$, the expected value of the random variable raised to the r th power. The first moment about the origin is the mean, while higher-order raw moments help characterize scale and overall magnitude. In contrast, the r th moment about the mean (central moment) is $\mathbb{E}[(X - \mu)^r]$, where $\mu = \mathbb{E}[X]$. Central moments describe variability and shape independent of location: the second central moment is the variance (dispersion), the third relates to skewness (asymmetry), and the fourth to kurtosis (tail heaviness).

Conceptually, a moment can be interpreted analogously to mass $\times$ distance in mechanics: if probability density is viewed as a distributed mass, then moments describe how that mass is distributed relative to a reference point (the origin or the mean). Just as physical moments quantify rotational tendency about a point, statistical moments quantify how probability mass is distributed and how strongly it spreads or tilts about a reference location.

1.2.1 Mean

The *mean* or expected value μ is the first moment about the origin and represents the center of mass of the probability density function for a population.

$$\mathbb{E}[X] = \mu = \int_{-\infty}^{\infty} x f_X(x) dx \quad (1.8)$$

For a constant c and function $g(x)$, the following rules apply:

$$\mathbb{E}[c] = c \quad (1.9)$$

$$\mathbb{E}[c \cdot g(x)] = c \cdot \mathbb{E}[g(x)] = c \cdot \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx \quad (1.10)$$

$$\mathbb{E}[g_1(x) \pm g_2(x)] = \mathbb{E}[g_1(x)] \pm \mathbb{E}[g_2(x)] \quad (1.11)$$

The corresponding sample statistic is the sample mean $\bar{x}$. It represents the average value of the data and the center of mass of the histogram.

$$\hat{\mu} = \bar{x} = \sum_{i=1}^N \frac{x_i}{N} \quad (1.12)$$

1.2.2 Median

The *median* represents an alternative measure to the mean if the frequency distribution is skewed, and therefore the mean is not very robust.

$$m = 50^{\text{th}} \text{ percentile} \quad (1.13)$$

$$\int_{-\infty}^m f_X(x) dx = F_X(m) = 0.5 \quad (1.14)$$

1.2.3 Mode

The *mode* describes the most frequently occurring value of the frequency distribution and is therefore the value of X that maximizes $f_X(x)$.

$$\left. \frac{df_X(x)}{dx} \right|_{\text{mode}} = 0 \quad \text{and} \quad \frac{d^2 f_X(x)}{dx^2} < 0 \quad (1.15)$$

A data sample can have no, one, or multiple modes.

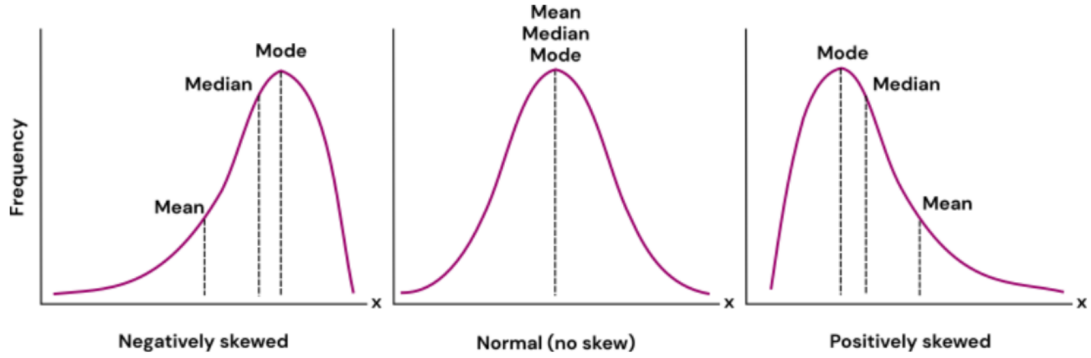


Figure 1.2. Visualization of the relation between the mean, median and mode for three different distributions: negatively skewed (left), normal (middle), and positively skewed (right). (Ledidi 2025)

1.2.4 Variance and Standard Deviation

The variability of data is measured by the *variance* σ_X^2 , which is the second moment about the mean. The *standard deviation* σ_X is the square root of the variance and has the same units as X .

$$\text{Var}[X] = \sigma_X^2 \quad (1.16)$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx \quad (1.17)$$

$$= \mathbb{E}[(X - \mu)^2] \quad (1.18)$$

$$= \mathbb{E}[X^2] - \mathbb{E}^2[X] \quad (1.19)$$

For the constants a , b , and c , the following rules apply:

$$\text{Var}[c \cdot X] = c^2 \cdot \text{Var}[X] \quad (1.20)$$

$$\text{Var}[a + b \cdot X] = b^2 \cdot \text{Var}[X] \quad (1.21)$$

$$\text{Var}[c] = 0 \quad (1.22)$$

The corresponding sample statistics are the sample variance s_X^2 and the sample standard deviation s_X . The left graph in Figure 1.3 visualizes two distributions of differing standard deviation.

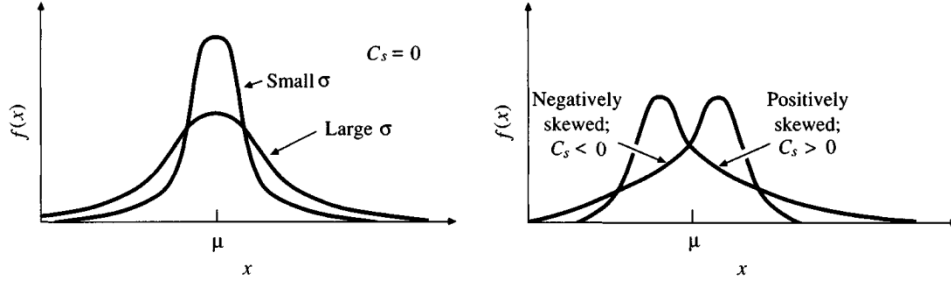


Figure 1.3. Visualization of different standard deviations σ_X (left) and coefficients of skewness C_s (right). (Chow et al. 1988, Fig. 11.3.1)

$$\hat{\sigma}_X^2 = s_X^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1} \quad (1.23)$$

$$s_X = \sqrt{s_X^2} \quad (1.24)$$

1.2.5 Coefficient of Variation

The *coefficient of variation* CV describes the data spread relative to the magnitude of the variable. The coefficient of variation is especially useful when comparing variability across datasets that have different units or very different average values, since it shows how large the spread is relative to the mean and therefore offers a dimensionless descriptor.

$$CV = \frac{s_X}{\bar{x}} \quad (1.25)$$

1.2.6 Interquartile Range

The *interquartile range* iqr represents an alternative measure to the variance for skewed distributions. It measures the range of the central 50 % of the data. The visualization of the interquartile range in relation to the box-and-whiskers plot is shown in Figure 1.5, while Figure 1.1 shows how the interquartile range and the cumulative frequency function are connected.

$$iqr = 75^{\text{th}} \text{ percentile} - 25^{\text{th}} \text{ percentile} \quad (1.26)$$

$$= Q_3 - Q_1 \quad (1.27)$$

1.2.7 Skewness

The *skewness* is a measure of the symmetry of the distribution around its central value and represents the third moment about the mean. The skewness is normally made dimensionless by dividing by σ^3 to give the coefficient of skewness γ .

$$\gamma = \frac{1}{\sigma^3} \mathbb{E}[(X - \mu)^3] \quad (1.28)$$

$$= \frac{1}{\sigma^3} \int_{-\infty}^{\infty} (x - \mu)^3 f_X(x) dx \quad (1.29)$$

We differentiate two cases, visualized in Figure 1.3 on the right:

- Long tail to the left: negative skewness, $\gamma < 0$
- Long tail to the right: positive skewness, $\gamma > 0$

The sample estimate is given by

$$\hat{\gamma} = C_s = \frac{N}{(N-1)(N-2)} \sum_{i=1}^N \frac{(x_i - \bar{x})^3}{s_X^3} \quad (1.30)$$

1.2.8 Quartile Skew Coefficient

The *quartile skew coefficient* q_s represents a resistant measure of skewness.

$$q_s = \frac{(75^{\text{th}} \text{ percentile} - 50^{\text{th}} \text{ percentile}) - (50^{\text{th}} \text{ percentile} - 25^{\text{th}} \text{ percentile})}{75^{\text{th}} \text{ percentile} - 25^{\text{th}} \text{ percentile}} \quad (1.31)$$

1.2.9 Kurtosis

The *kurtosis* describes the peakedness or flatness of a distribution and represents the fourth standardized moment about the mean.

$$\text{Kurt}[X] = K \quad (1.32)$$

$$= \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] \quad (1.33)$$

$$= \frac{1}{\sigma_X^4} \int_{-\infty}^{\infty} (x - \bar{x})^4 f_X(x) dx \quad (1.34)$$

The *kurtosis* of the normal distribution is $K = 3$. We can therefore define the *excess kurtosis* for a sample data C_c by subtracting three from the sample kurtosis. Samples with a negative excess kurtosis are therefore flatter than the normal distribution and vice versa, as visualized in Figure 1.4.

$$C_c = \frac{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^4}{s_X^4} - 3 \quad (1.35)$$

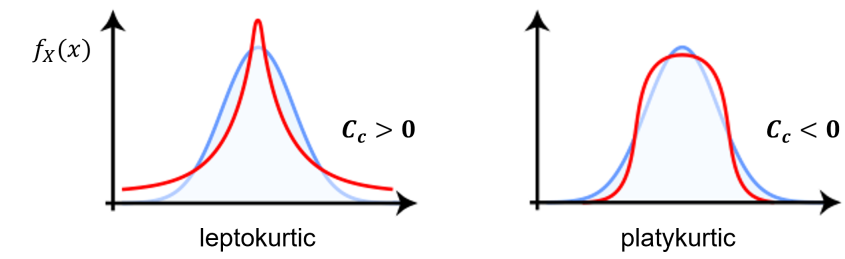


Figure 1.4. Visualization of a leptokurtic ($C_c > 0$) and a platykurtic ($C_c < 0$) distribution (red) compared to the normal distribution (blue) with $C_c = 0$. (adapted from Wikipedia 2025)

1.2.10 Box-and-Whisker Plot

The box-and-whisker plot is a helpful visualization to display the locality and spread of a frequency distribution. It is especially useful when comparing multiple distributions with each other. Figure 1.5 shows the meaning of the different components of the box-and-whisker plot. Additionally, the connection to the cumulative frequency function is visualized in Figure 1.1.

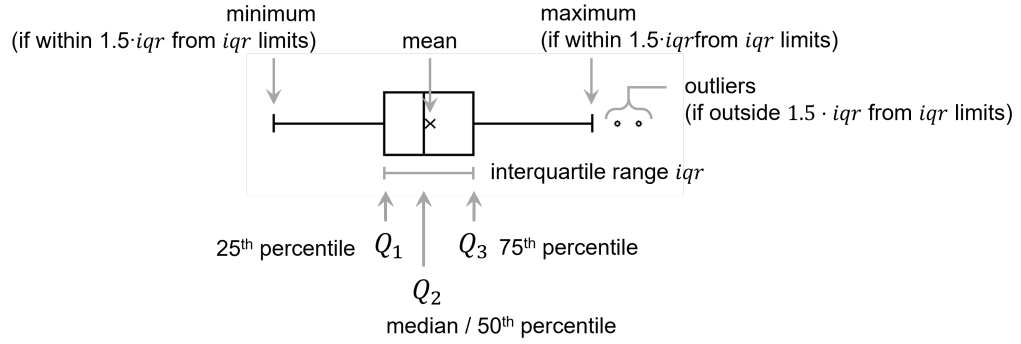


Figure 1.5. Explanation of the components of the box-and-whisker plot.

1.3 Probability Distributions and Properties

There are many different models for probability distributions available. The model choice depends on the type of variable (e.g. positive, discrete, or continuous) and what the variable is measuring (which hydrologic variable, extremes, etc.).

1.3.1 Uniform Distribution

A uniform distribution is a probability distribution in which every outcome in a given range is equally likely. For a continuous uniform distribution, any number between two bounds $[a, b]$ has the same probability density (see Fig. 1.6). It is useful for modeling completely random scenarios without bias, such as rolling a fair die, generating random numbers, or initializing parameters in simulations and algorithms where no preference is assumed.

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b \end{cases} \quad (1.36)$$

$$F_X(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases} \quad (1.37)$$

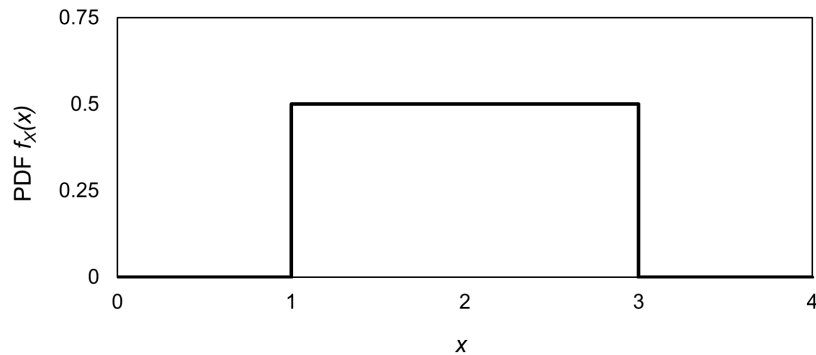


Figure 1.6. Visualization of the continuous uniform distribution function with bounds $a = 1$ and $b = 3$. The resulting probability density over the segment between the bounds is 0.5.

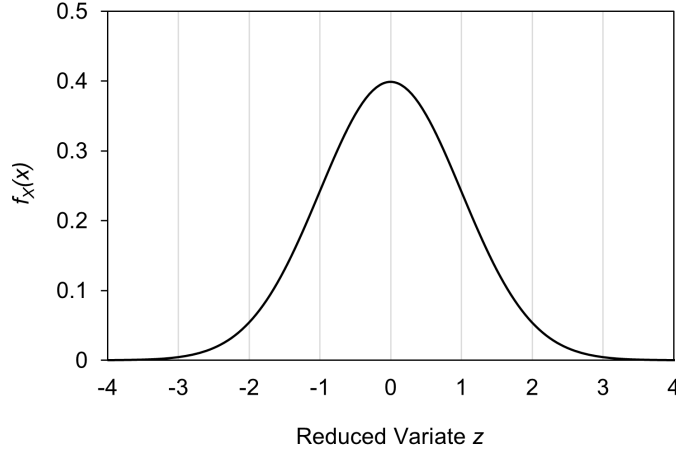


Figure 1.7. Visualization of the standard normal distribution $\mathcal{N}(0, 1)$.

1.3.2 Normal or Gaussian Distribution

The normal or Gaussian distribution arises from the central limit theorem, which states that if a sequence of random variables X_i are independently and identically distributed, then the distribution of the sum of such random variables tends towards the normal distribution as it becomes large. Hydrologic variables, such as annual precipitation, calculated as the sum of the effects of many independent events tend to follow the normal distribution. The main limitations of the normal distribution for describing hydrologic variables are that it varies over a continuous range $[-\infty, \infty]$, while most hydrologic variables are nonnegative, and that it is symmetric about the mean, while hydrologic data tend to be skewed. An possible application for hydrological modeling is therefore the modeling of errors.

The normal distribution is a 2-parameter distribution written as $\mathcal{N}(\mu, \sigma^2)$, where μ is the location and σ^2 the scale parameter.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right] \quad (1.38)$$

The standard normal distribution $\mathcal{N}(0, 1)$ describes the special case of the normal distribution with location $\mu = 0$ and scale $\sigma^2 = 1$ (visualized in Fig. 1.7). For every normal distribution, the reduced variate z as defined below in Equation 1.39 follows a standard normal distribution.

$$z = \frac{x - \mu}{\sigma} \sim \mathcal{N}(0, 1) \quad (1.39)$$

The cumulative distribution function of the standard normal distribution, often denoted $\Phi(z)$, is defined as the integral in Equation 1.40 and cannot be expressed in terms of elementary functions. Therefore, tables provide $\Phi(z) = F_X(x) = \mathbb{P}(X \leq x)$ that relate the reduced variate z to $\Phi(z)$. You find an example of a lookup table in Table A.1 in the Appendix.

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt \quad (1.40)$$

1.3.3 Lognormal Distribution

A variable X has a lognormal distribution, if its natural logarithm $Y = \ln(X)$ is Gaussian distributed. The lognormal distribution is a 2-parameter distribution described as $\mathcal{L}(\mu, \sigma^2)$, where the location μ and σ^2 are the statistics of the normally distributed $\ln(X)$. The corresponding expression of the probability density function $f_X(x)$ of the lognormal distribution is provided in Equation 1.41 and visualized in Figure 1.8.

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{\ln(x) - \mu}{\sigma} \right)^2 \right] \quad (1.41)$$

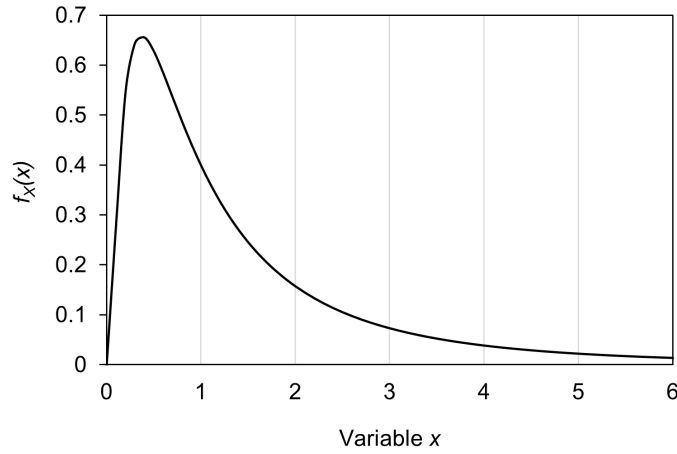


Figure 1.8. Visualization of the lognormal distribution $\mathcal{L}(0, 1)$.

In hydrology, the lognormal distribution finds application for the modeling of the hydraulic conductivity. The lognormal distribution has the advantages over the normal distribution that it is bounded ($x > 0$) and that the log transformation tends to reduce the positive skewness commonly found in hydrologic data, because taking logarithms reduces large numbers proportionately more than it does small numbers. Some limitations of the lognormal distribution are that it has only two parameters and that it requires the logarithms of the data to be symmetric about their mean.

1.3.4 Exponential Distribution

Some sequences of hydrologic events, such as the occurrence of precipitation, may be considered Poisson processes, in which events occur instantaneously and independently on a time horizon, or along a line. The time between such events, or inter arrival time, is described by the exponential distribution whose parameter λ is the mean rate of occurrence of the events. The exponential distribution is used to describe the inter arrival times of random shocks to hydrologic systems, such as slugs of polluted runoff entering streams as rainfall washes the pollutants off the land surface. The advantage of the exponential distribution is that it is easy to estimate λ from observed data (only one parameter is needed) and the exponential distribution lends itself well to theoretical studies, such as a probability model for the linear reservoir ($\lambda = 1/k$, where k is the storage constant in the linear reservoir). Its disadvantage is that it requires the occurrence of each event to be completely independent of its neighbors, which may not be a valid assumption for the process under study.

$$f_X(x) = \lambda e^{-\lambda x} \quad (1.42)$$

$$F_X(x) = 1 - e^{-\lambda x} \quad (1.43)$$

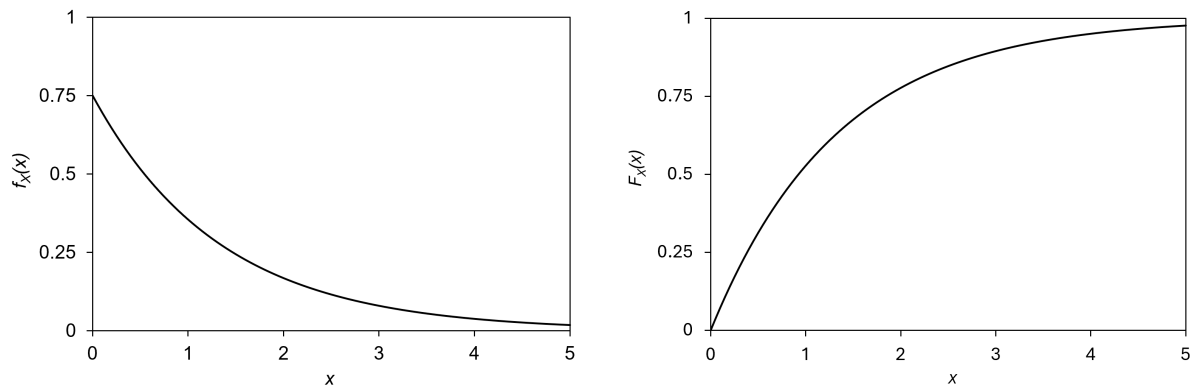


Figure 1.9. Probability density function $f_X(x)$ (left) and cumulative distribution function $F_X(x)$ (right) of the exponential distribution with $\lambda = 0.75$.

Parameter estimation by method of moments of maximum likelihood:

$$\bar{x} = \frac{1}{\hat{\lambda}} \quad (1.44)$$

$$s_X^2 = \frac{1}{\hat{\lambda}^2} \quad (1.45)$$

1.3.5 2-Parameter Gamma Distribution

The 2-parameter Gamma distribution $\Gamma(k, \lambda)$ models the sum of k (shape parameter) exponentially distributed independent random variables with parameter λ (scale parameter). Equation 1.46 describes the probability density function $f_X(x)$, where $\Gamma(\cdot)$ describes the 1-parameter Gamma distribution.

$$f_X(x) = \frac{x^{k-1} \lambda^k e^{-\lambda x}}{\Gamma(k)} \quad x \geq 0, k > 0, \lambda > 0 \quad (1.46)$$

The 2-parameter Gamma distribution is useful for the application to storm rainfall depth measurements and travel times in a catchment. It has the advantage of being a flexible distribution for strictly positive variables and can describe skewed variables without transformation. In cases where a lower bound of zero is not appropriate, the 2-parameter Gamma distribution is disadvantageous.

Parameter estimation by the method of moments:

$$\bar{x} = \frac{\hat{k}}{\hat{\lambda}} \quad (1.47)$$

$$s_X^2 = \frac{\hat{k}}{\hat{\lambda}^2} \quad (1.48)$$

$$C_s = \frac{2}{\sqrt{\hat{k}}} \quad (1.49)$$

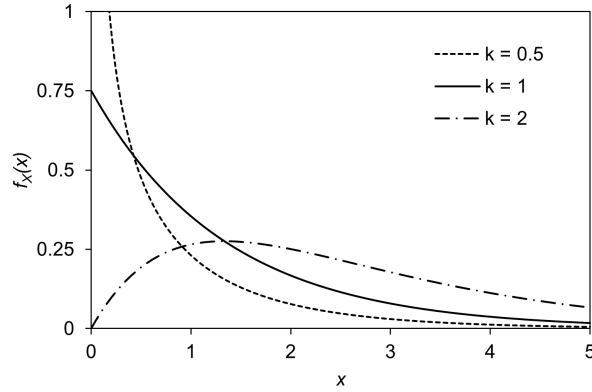


Figure 1.10. Visualization of the 2-parameter Gamma distribution with $\lambda = 0.75$, and $k = 0.5, 1$ and 2 .

1.3.6 Generalized Extreme Value Distribution

Distributions of the extreme values selected from sets of samples of any probability distribution have been shown to converge to one of three forms of extreme value distributions, called Types I, II and III, respectively, when the number of selected extreme values is large. The properties of the three limiting forms were further developed by Gumbel for the Extreme Value Type I (EV-I) distribution, Fréchet for the Extreme Value Type II (EV-II), and Weibull for the Extreme Value Type III (EV-III). These three forms were also shown to be special cases of the *Generalized Extreme Value distribution GEV*. Its probability and cumulative distribution function are given in Equations 1.50 and 1.51. Note that the parameters $-\infty < \mu < \infty$, $\sigma > 0$ and $-\infty < \xi < \infty$ refer to the location, scale, and shape parameters, respectively. They do not represent the mean or standard deviation of the population.

$$f_X(x) = \frac{1}{\sigma} \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-\frac{1}{\xi} - 1} \cdot \exp \left\{ - \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-\frac{1}{\xi}} \right\} \quad \text{for } \xi \neq 0 \quad (1.50)$$

$$F_X(x) = \exp \left[- \left(1 - \xi \cdot \frac{x - \mu}{\sigma} \right)^{1/\xi} \right] \quad \text{for } \xi \neq 0 \quad (1.51)$$

The three types are differentiated by the shape parameter ξ (see Fig. 1.11):

- Gumbel distribution / EV-I: $\xi = 0$
- Fréchet distribution / EV-II: $\xi > 0$, supported for $x > \mu - \sigma/\xi$
- Weibull distribution / EV-III: $\xi < 0$, supported for $x < \mu - \sigma/\xi$

You can find information on the parameter estimation for the GEV distribution for example in the textbook *Statistical Methods in the Atmospheric Sciences* (Wilks 2011).

For the case of $\xi = 0$ of the *Gumbel distribution* / EV-I, the equations simplify to the two parameter distribution given in Equations 1.52 and 1.53. This distribution finds wide application in the modeling of annual maximum discharge.

$$f_X(x) = \frac{1}{\sigma} \exp \left[-\frac{x - \mu}{\sigma} - \exp \left(-\frac{x - \mu}{\sigma} \right) \right] \quad (1.52)$$

$$F_X(x) = \exp \left[-\exp \left(-\frac{x - \mu}{\sigma} \right) \right] \quad (1.53)$$

Parameter estimation by the method of moments:

$$\bar{x} = \hat{\mu} + 0.5772 \cdot \hat{\sigma} \quad (1.54)$$

$$s_X = \hat{\sigma} \cdot \frac{\pi}{\sqrt{6}} \quad (1.55)$$

The *Weibull distribution* / EV-III can be simplified to a two parameter distribution as well by fixing its support to $x \geq 0$. Therefore, the location parameter can be removed and the distribution is described by the scale parameter β and the shape parameter k . The equation of the probability and cumulative distribution functions are provided in Equations 1.56 and 1.57. This distribution finds wide application in the modeling of low flows, e.g. annual minimum discharge. For information on parameter estimation, consult the *Statistical Methods in the Atmospheric Sciences* (Wilks 2011).

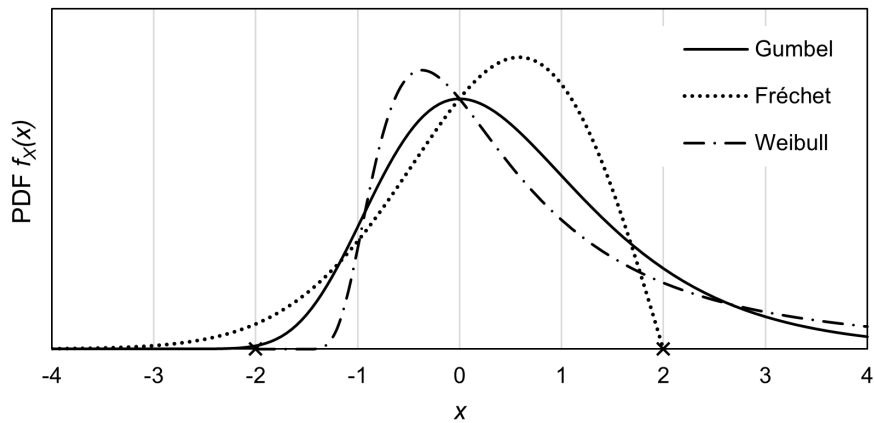


Figure 1.11. Visualization of the generalized extreme value distribution GEV for the three cases Gumbel / EV-I ($\xi = 0$), Fréchet / EV-II ($\xi = -0.5$), and Weibull / EV-III ($\xi = 0.5$). The scale and shape parameter are the same for all the distributions with $\mu = 0$ and $\sigma = 1$. The crosses represent the end points of the support for the Fréchet and Weibull distributions.

$$f_X(x) = \left(\frac{k}{\beta}\right) \left(\frac{x}{\beta}\right)^{k-1} \exp \left[-\left(\frac{x}{\beta}\right)^k \right] \quad \text{for } x > 0; \beta, k > 0 \quad (1.56)$$

$$F_X(x) = 1 - \exp \left[-\left(\frac{x}{\beta}\right)^k \right] \quad \text{for } x > 0; \beta, k > 0 \quad (1.57)$$

1.3.7 Bootstrap Confidence Intervals

Bootstrapping is a general approach used to address many estimation problems in stochastic hydrology without assuming a specific theoretical distribution. Instead, it relies on the observed data itself by repeatedly sampling with replacement—typically hundreds to thousands of times—so that each observation can be selected multiple times in a resample. Each resample has the same size as the original dataset (e.g., 25 observations), and for each one, a statistic such as the mean, median, or standard deviation is computed. This process produces many estimates of the statistic, forming an empirical sampling distribution. Confidence intervals are then derived by sorting these estimates and selecting appropriate percentiles from the resulting distribution.

1.4 Parameter Estimation

A probability distribution is a function representing the probability of occurrence of a random variable. By fitting a distribution to a set of hydrologic data, a great deal of the probabilistic information in the sample can be compactly summarized in the function and its associated parameters. Fitting distributions can be accomplished by the *method of moments* or the *method of maximum likelihood*.

The goal is to find the best parameters of the chosen probability distribution function $\hat{\theta}$ based on the sample data and thus to estimate the unknown population parameter θ . Note that the hat $\hat{\cdot}$ above a parameter indicates that it represents an estimator of the population parameter. This makes the estimator itself a random variable, since it varies from sample to sample.

1.4.1 Method of Moments

The *method of moments* considers that good estimates of the parameters of a probability distribution are those for which moments of the probability density function are equal to the corresponding moments of the sample data. For a distribution with m parameters, equate the first m moments of the distribution to the first m sample moments.

Steps for parameter estimation:

1. Derive the theoretical moments as functions of the parameters of the probability density function PDF. This is generally provided.
2. Estimate the sample moments from the data.
3. Equate the theoretical and sample moments and solve for the PDF parameters.

Under regular conditions, the method of moments is consistent, while it is not guaranteed to be unbiased or sufficient. In general, the method is not efficient.

1.4.2 Method of Maximum Likelihood

The *method of maximum likelihood* reasons that the best value of a parameter of a probability distribution should be that value which maximizes the likelihood or joint probability of occurrence of the observed sample. Given a sample of independent and identically distributed observations $x_1, x_2, \dots, x_n$, the value of the probability density for $X = x_i$ is $f(x_i)$.

Steps for parameter estimation:

1. Construct the likelihood function $\mathcal{L}$:

$$\mathcal{L} = f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n | \underline{\theta}) \quad (1.58)$$

$$= f_{X_1}(x_1 | \underline{\theta}) \cdot f_{X_2}(x_2 | \underline{\theta}) \cdot \dots \cdot f_{X_n}(x_n | \underline{\theta}) \quad (1.59)$$

$$= \prod_{i=1}^n f_X(x_i | \underline{\theta}) \quad (1.60)$$

2. Find the optimal parameter set $\underline{\theta}$ where the likelihood function $\mathcal{L}$ is maximized:

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad (1.61)$$

$$\frac{\partial^2 \mathcal{L}}{\partial \theta^2} < 0 \quad (1.62)$$

The method of maximum likelihood is asymptotically unbiased (as n gets large), more efficient than the method of moments (smaller mean square error MSE), consistent, and very often sufficient (but not guaranteed).

1.4.3 Properties of Estimators

Methods for parameter estimations can have several properties:

- *Unbiasedness*: The sample statistic $\hat{\theta}$ is a unbiased estimator of the true population if its expected value $\mathbb{E}[\hat{\theta}]$ is equal to true θ .

$$\mathbb{E}[\hat{\theta}] = \theta \quad (1.63)$$

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta \quad (1.64)$$

- *Consistency*: The probability of the bias between the sample and population parameter being larger than a small number ε converges to zero for a sample size n going to infinity.

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta} - \theta| > \varepsilon) = 0 \quad (1.65)$$

- *Efficiency*: An efficient estimator uses the data in the best possible way to give an estimate with the smallest possible margin of error. It therefore both minimizes the bias and the variance, to obtain the estimator with the lowest mean squared error MSE.

$$\min[\text{MSE}(\hat{\theta})] = \min[\text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2] \quad (1.66)$$

$$\min \mathbb{E}[(\hat{\theta} - \theta)^2] = \min [\text{Var}(\hat{\theta}) + (\mathbb{E}[\hat{\theta}] - \theta)^2] \quad (1.67)$$

- *Sufficiency*: An estimator is sufficient if it captures all the information in the data about the parameter. A sufficient estimator doesn't require any additional data for estimation.

2 Hypothesis Testing

2.1 Definitions

Hypothesis testing provides a rigorous statistical tool to help in decision making. The following list provides the basic definitions for hypothesis testing.

- The hypothesis being tested is called the *Null Hypothesis* H_0 . It is usually a hypothesis of no change or no difference.
- The *Alternative Hypothesis* H_1 describes some departure from H_0 that might be expected.
- The *test statistic* T is a random variable which is defined as some function of the sample being tested. Test statistics are always dimensionless. The probability distribution of T , given H_0 , is a function of sample size only.
- In a *one-sided hypothesis test*, H_1 is a departure from H_0 only in one direction.
- In a *two-sided hypothesis test*, H_1 is a departure from H_0 in either direction.
- *Decision rule for the test*: Reject H_0 if $T < T_L$ or if $T > T_U$
 T_L and T_U are the lower and upper critical values of the test statistic T . The critical value is not dependent on the observations but is a function of the significance level α , the sampling distribution of the test (e.g., z -test, t -test, χ^2 -test or F -test), the degrees of freedom df and whether a one- or two-sided test is employed.
- The probability of rejecting H_0 assuming H_0 is true $\mathbb{P}(\text{reject } H_0 | H_0 \text{ true})$ is called the significance level α .
- The *attained significance level* p describes the smallest level α for which H_0 would be rejected given the observed value of T . If $p < \alpha$, then reject H_0 . The smaller the value of p , the stronger the evidence against H_0 (see Fig. 2.1).
- The *power of a test* describes the probability that H_0 will be rejected given a particular kind and magnitude of departure from H_0 . It is equal to $1 - \mathbb{P}(\text{type II error}) = 1 - \beta$.

The possible outcomes of a hypothesis test and their probabilities are summarized in Table 2.1:

Table 2.1. Possible outcomes of a hypothesis test and their probabilities.

Decision	States of Nature	
	H_0 is true	H_0 is false
Do not reject H_0	Correct decision. The probability of this outcome is $1 - \alpha$.	Type II error. The probability of this outcome is β .
Reject H_0	Type I error. The probability of this outcome is α .	Correct decision. The probability of this outcome is $1 - \beta$ (power of the test).

2.1.1 Structure of Hypothesis Tests

A hypothesis test is performed by following the structure outlined below:

1. Choose the appropriate test.
2. Establish the null H_0 and alternative hypothesis H_1 .
3. Decide on an acceptable error rate α .
4. Compute the test statistic from the data, following the procedure for the chosen test.
5. Compute the p -value as a function of the observed test statistic.
 Alternatively, compute the critical values of the test statistic T_L and T_U based on the chosen significance level α .

6. Reject the null hypothesis H_0 if $p \leq \alpha$.

Alternatively, reject H_0 if the test statistic $T < T_L$ or $T > T_U$, where T_L and T_U are the lower and upper critical values of the test statistic. Figure 2.1 visualizes the two cases where the null hypothesis H_0 is accepted and rejected for a standard normally distributed test statistic at the $\alpha = 5\%$ significance level.

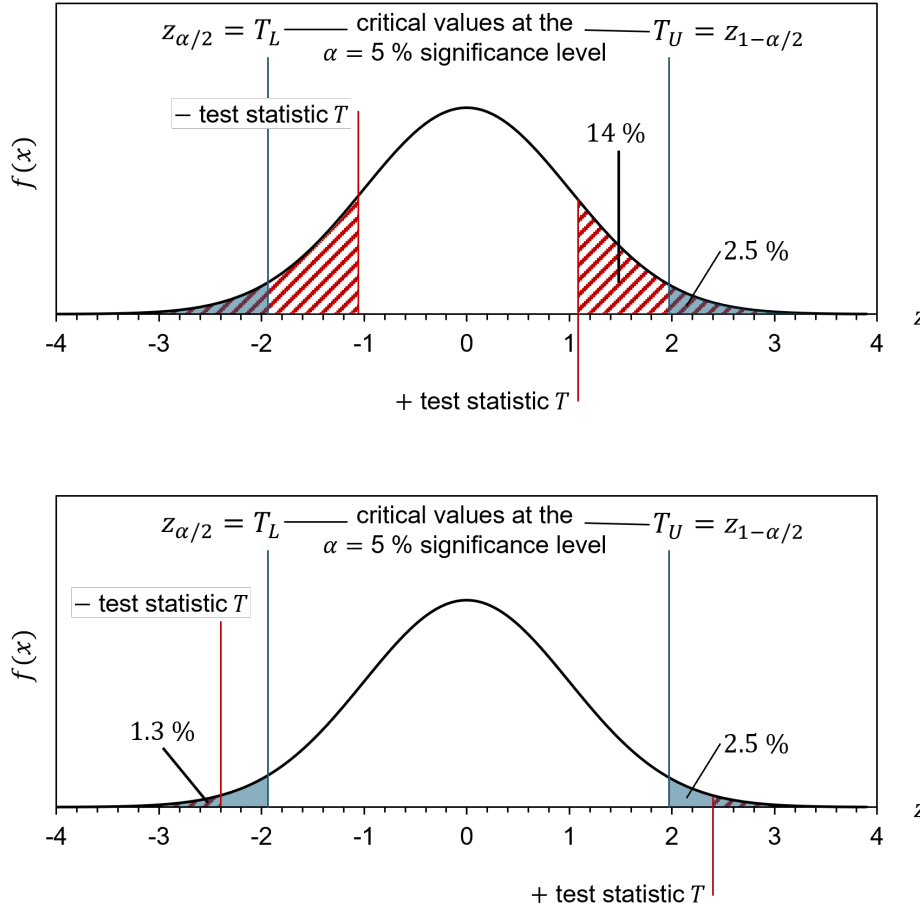


Figure 2.1. Visualization of a hypothesis test where the test statistic T follows a standard normal distribution and a two-tailed hypothesis test at the $\alpha = 5\%$ significance level is performed. The upper plot represents a case where the null hypothesis H_0 is accepted with $p = 28\% > \alpha = 5\%$. The lower plot represents a case where the null hypothesis is rejected with $p = 2.6\% \leq \alpha = 5\%$.

2.1.2 Parametric and Nonparametric Tests

A *parametric* test comprises any hypothesis test that includes the specification of the type of data. In contrast, a *nonparametric* or distribution free test includes no assumption about the type of data distribution. If the distributional assumptions of a parametric test are correct, the power (for a given α) of the parametric test is greater than that of a nonparametric test. Conversely, if the distribution departs substantially from the assumed distribution, the nonparametric test can be much more powerful than the parametric test. Therefore, if the a priori distribution characteristics of the data are known, use a parametric hypothesis test. For parametric tests that assume a normal distribution, the significance levels are reasonably accurate regardless of the actual distribution of the data. However, the power of the test may be severely diminished if the data are truly nonnormal.

Many hydrologic data sets are positively skewed and have variability proportional to the magnitude of the data. Logarithmic or power transformations can be effective in making these distributions approximately symmetrical and more nearly normal, so that parametric tests can be used. Transformations do not necessarily eliminate the "heavy tails" of a distribution.

If possible, try both parametric and nonparametric tests in any given problem. If the two analyses give similar results, this should add confidence to the conclusion. If they do not, this should lead to more data analysis to understand the cause of the discrepancy.

2.1.3 Permutation Tests

Permutation tests are non-parametric, exact hypothesis tests that determine statistical significance by looking at the possible rearrangements of the observed data to build a null distribution, rather than assuming a normal distribution. They are ideal for small samples, skewed data, or when data distribution is unknown, making them a powerful alternative to t-tests.

The test repeatedly shuffles the data, recalculates the statistic for each shuffle, and builds an empirical reference distribution. Therefore, this type of test is more computationally intensive than the tests without reshuffling. The p -value is then computed as the proportion of outcomes that are equal to or more extreme than the one obtained from data.

2.2 Common Hypothesis Tests

2.2.1 Test of Hypothesis about the Sample Mean - Parametric

For the test about the sample mean, the following null and alternative hypothesis hold:

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0$$

The sample is of size n with mean value $\bar{X}$ and standard deviation s . The assumptions are that the observations are normal, independent, and identically distributed. If the sample size $n > 30$, the normality assumption can be relaxed and the significance level of the test will remain accurate, but the power may be reduced as compared to a nonparametric approach such as the median test.

One situation in which the test would be used is where some independent analysis indicated that the population mean is μ_0 . The test then becomes a test of that independent analysis.

The test statistic T is computed as

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \quad (2.1)$$

Decision rule:

$$\text{Reject } H_0 \text{ if: } |T| > t_{1-\alpha/2, n-1}$$

where $t_{1-\alpha/2, n-1}$ is the $1 - \alpha/2$ quantile of the Student's t distribution with $n - 1$ degrees of freedom. A lookup table of the Student's t distribution is found in Table A.2 in the Appendix.

2.2.2 Test of Hypothesis for the Population Median - Nonparametric

For the test for the population median, the following null and alternative hypothesis hold:

$$H_0 : \text{population median} = M$$

$$H_1 : \text{population median} \neq M$$

The assumptions are that the data consists of n independent and identically distributed samples draw from a single distribution. The test is more powerful than the test for the population mean in cases where the distribution departs substantially from the normal.

The test statistic T is the number of observations that are less than or equal to M (assumed population median).

Decision rule (applies for $n > 20$):

$$\text{Reject } H_0 \text{ if: } \left| T - \frac{n}{2} \right| \geq z_{1-\alpha/2} \cdot \frac{\sqrt{n}}{2}$$

where $z_{1-\alpha/2}$ represents the z value of the standard normal distribution for which the cumulative area under the curve is $\Phi(z) = 1 - \frac{\alpha}{2}$. A lookup table of the standard normal distribution is provided in Table A.1 in the Appendix.

2.3 Paired Comparisons

Paired comparisons occur when there is a logical pairing of observations between two groups. The question is whether there is a tendency for one group to have higher values than the other. Differences between the groups are seen clearly when tests are performed on the differences between data pairs. For paired observations (x_i, y_i) , $i = 1, 2, \dots, n$, the null hypothesis that x_i and y_i are from the same population implies that the mean or median of their differences $D_i = x_i - y_i$ is zero.

2.3.1 Signed-Rank Test - Nonparametric

For the signed-rank test, the null hypothesis assumes that the x 's and y 's come from the same distribution:

$$H_0 : \text{median}[D] = 0$$

The alternative hypothesis is one of three statements:

$$H_1 : \text{median}[D] \neq 0 \quad (2\text{-sided test; } x \text{ might be larger or smaller than } y)$$

$$H_2 : \text{median}[D] > 0 \quad (1\text{-sided test; } x \text{ is larger than } y)$$

$$H_3 : \text{median}[D] < 0 \quad (1\text{-sided test; } x \text{ is smaller than } y)$$

There is no requirement that the distributions of x and y be normal or that they even be symmetric. The test is more powerful than parametric tests when the data are not normally distributed.

Computational steps:

1. Eliminate all $D_i = 0$ from the data set (the cases where $x_i = y_i$).
2. Define n as the number of nonzero D_i values.
3. Rank the D_i according to their absolute values. Largest $|D_i|$ has rank n , the smallest $|D_i|$ has rank 1. If there are ties, assign them an average rank.
4. Let R^+ = sum of the ranks corresponding to $D_i > 0$.
5. Let R^- = sum of the ranks corresponding to $D_i < 0$.
6. Compute the test statistic W^+ , the minimum of $\{R^+, R^-\}$.
7. Large sample approximation if $n > 15$:
For the sample size n , compute the theoretical mean and standard deviation of W^+ under H_0 . They are

$$\mu_{W^+} = \frac{n(n+1)}{4} \quad \sigma_{W^+} = \sqrt{\frac{n(n+1)(2n+1)}{24}} \quad (2.2)$$

8. Compute the standardized form of the test statistic Z_{sr+}

$$Z_{sr+} = \begin{cases} \frac{W^+ - 0.5 - \mu_{W^+}}{\sigma_{W^+}} & \text{if } W^+ > \mu_{W^+} \\ 0 & \text{if } W^+ = \mu_{W^+} \\ \frac{W^+ + 0.5 - \mu_{W^+}}{\sigma_{W^+}} & \text{if } W^+ < \mu_{W^+} \end{cases} \quad (2.3)$$

9. Decision Rule:

$$H_1 \rightarrow \text{Reject } H_0 \text{ if: } |Z_{sr+}| \geq z_{1-\alpha/2}$$

$$H_2 \rightarrow \text{Reject } H_0 \text{ if: } Z_{sr+} \geq z_{1-\alpha}$$

$$H_3 \rightarrow \text{Reject } H_0 \text{ if: } Z_{sr+} \leq -z_{1-\alpha}$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ point on the standard normal distribution. A lookup table of the standard normal distribution is provided in Table A.1 in the Appendix.

For small sample sizes $n \leq 15$, the large sample approximation is imprecise and the exact form should be considered. For the test statistic W^+ , the following decision rules apply:

$$H_1 \rightarrow \text{Reject } H_0 \text{ if: } W^+ \geq x_{\alpha/2, n} \text{ or } W^+ \leq x'_{\alpha/2, n}$$

$$H_2 \rightarrow \text{Reject } H_0 \text{ if: } W^+ \geq x_{\alpha, n}$$

$$H_3 \rightarrow \text{Reject } H_0 \text{ if: } W^+ \leq x'_{\alpha, n}$$

A Table of the critical test statistics x and x' can be found in Figure A.1 in the Appendix.

2.3.2 Paired t -Test - Parametric

The paired t -test is the most widely used test to evaluate matched pairs of data. However, it assumes that the differences D_i follow a normal distribution.

$H_0 : \mu_x = \mu_y$ ($D_i = x_i - y_i$ is normally distributed with mean zero)

Computational steps:

1. Compute $D_i = x_i - y_i$.
2. Compute the mean difference $\bar{D}$, the average of all D_i .
3. Compute the sample variance

$$S^2 = \sum_{i=1}^n \frac{(D_i - \bar{D})^2}{n-1} \quad (2.4)$$

4. Compute the standardized test statistic t_p

$$t_p = \frac{\bar{D}}{\sqrt{S^2/n}} \quad (2.5)$$

5. Decision rule:

$$\begin{aligned} H_1 : \mu_x \neq \mu_y &\rightarrow \text{Reject } H_0 \text{ if: } |t_p| > t_{1-\alpha/2, n-1} \\ H_2 : \mu_x > \mu_y &\rightarrow \text{Reject } H_0 \text{ if: } t_p > t_{1-\alpha, n-1} \\ H_3 : \mu_x < \mu_y &\rightarrow \text{Reject } H_0 \text{ if: } t_p < -t_{1-\alpha, n-1} \end{aligned}$$

where $t_{1-\alpha/2, n-1}$ is the $1 - \alpha/2$ quantile of a Student's t distribution with $n - 1$ degrees of freedom. A lookup table of the t distribution can be found in the Table A.2 in the Appendix.

2.3.3 Paired Permutation Test

The permutation test for paired differences can be used to test whether the mean differences equal zero regardless of the distribution of the differences. It performs better than the paired t test in the presence of asymmetry.

Computational steps:

1. Compute all the differences between the data pairs $D_i = x_i - y_i$.
2. Randomly assign an algebraic sign (+/-) to each of the observed differences.
3. Compute the mean difference $\bar{D}$.
4. Construct the distribution of the test statistic $\bar{D}$ by repeating steps 2 and 3 a few thousand times.
5. The observed mean difference is compared to the distribution of the test statistic. The p -value is obtained from the proportion of the distribution that is outside (for a 2-sided test) or above / below (for a 1-sided test) the observed mean difference.

2.4 Comparison between Two Independent Data Groups

Tests to compare two independent groups of data are performed to determine whether one group tends to contain larger values than the other. The data are independent in the sense that there are no pairings of data between observation 1 of group A and observation 1 of group B, etc. An example of such a situation is a study in which chloride concentrations at base flow during winter are collected in a random sample of stream reaches in rural basins and a random sample of reaches in basins with substantial urban land use. Even if the number of samples in each group were equal, there is no natural pairing of the data from the two groups, and the paired sample methods are not appropriate.

2.4.1 Rank-Sum Test - Nonparametric

The rank-sum test, the Wilcoxon rank-sum test, the Mann-Whitney test, and the Wilcoxon-Mann-Whitney rank sum test are all the same test and will be referred to by the name in the title in these lecture notes. No assumptions are made about the how the data are distributed. The test's results are not affected by monotonic transformations of the data (such as log transformation). The null hypothesis assumes that both sample groups x (smaller size n) and y (larger size m) are identically distributed.

$$H_0 : \mathbb{P}[x > y] = 0.5$$

The alternative hypothesis is one of three statements:

$$H_1 : \mathbb{P}[x > y] \neq 0.5 \quad (2\text{-sided test; } x \text{ might be larger or smaller than } y)$$

$$H_2 : \mathbb{P}[x > y] > 0.5 \quad (1\text{-sided test; } x \text{ is expected to be larger than } y)$$

$$H_3 : \mathbb{P}[x > y] < 0.5 \quad (1\text{-sided test; } x \text{ is expected to be smaller than } y)$$

Computational steps:

1. Rank the data in ascending order from 1 (smallest) to N (largest), where $N = n + m$. In case of ties (equal data values), use the average of the ranks for the tied values.
2. Compute the test statistic W as the sum of the ranks of the n observations in the smaller group.
3. For large samples, where both groups are > 10 , the following approximation can be applied:
For the sample size, compute the theoretical mean and standard deviation of W under H_0 . They are

$$\mu = \frac{n(N+1)}{2} \quad \sigma = \sqrt{\frac{n \cdot m(N+1)}{12}} \quad (2.6)$$

If your data set contains ties (at least two data points of the same value), use the equation below instead for the computation of σ :

$$\sigma = \sqrt{\frac{n \cdot m}{N(N-1)} \cdot \sum_{k=1}^N R_k^2 - \frac{n \cdot m(N+1)^2}{4(N-1)}} \quad (2.7)$$

where R_k is the rank of observation k and therefore $\sum_{k=1}^N R_k^2$ represents the sum of all N squared ranks. The standardized form of the test statistic Z_{rs} is computed as

$$Z_{rs} = \begin{cases} \frac{W-0.5-\mu}{\sigma} & \text{if } W > \mu \\ 0 & \text{if } W = \mu \\ \frac{W+0.5-\mu}{\sigma} & \text{if } W < \mu \end{cases} \quad (2.8)$$

4. Decision rule:

$$H_1 \rightarrow \text{Reject } H_0 \text{ if: } |Z_{rs}| > z_{1-\alpha/2}$$

$$H_2 \rightarrow \text{Reject } H_0 \text{ if: } Z_{rs} > z_{1-\alpha}$$

$$H_3 \rightarrow \text{Reject } H_0 \text{ if: } Z_{rs} < -z_{1-\alpha}$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ point on the standard normal distribution. A lookup table of the standard normal distribution is provided in Table A.1 in the Appendix.

For small sample sizes $n, m < 10$, the large sample approximation is imprecise and the exact form should be considered. For the test statistic W , the following decision rules apply, where x and x^* represent the critical test statistics:

$$H_1 \rightarrow \text{Reject } H_0 \text{ if: } W \geq x_{\alpha/2, n, m} \text{ or } W \leq x_{\alpha/2, n, m}^*$$

$$H_2 \rightarrow \text{Reject } H_0 \text{ if: } W \geq x_{\alpha, n, m}$$

$$H_3 \rightarrow \text{Reject } H_0 \text{ if: } W \leq x_{\alpha, n, m}^*$$

A lookup table of critical test statistics x and x^* can be found in Figure A.2 in the Appendix.

2.4.2 Two-Sample t Test - Parametric

The t test is the most widely used method for comparing two independent groups of data. There are two issues that make the t test less applicable for general use than the sum-rank test. (1) it lacks power when applied to nonnormal data and (2) it is powerful for the case where the two groups differ by an additive constant but is not powerful for the case where the two cases differ by a multiplicative factor. In general, the rank-sum test should be used instead of the t test when the data is substantially skewed or nonnormal.

The null hypothesis H_0 assumes that both data groups x and y have the same mean.

$$H_0 : \mu_x = \mu_y$$

Computational steps:

1. Compute the sample means and sample standard deviations of the data in both groups $\bar{x}$, s_x and $\bar{y}$, s_y , where x denotes the first group (sample size n) and y denotes the second group (sample size m).
2. Compute the degrees of freedom df :

$$df = n + m - 2 \quad (2.9)$$

3. The test statistic t is computed as

$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n} + \frac{1}{m}}} \quad (2.10)$$

$$\text{with } s = \sqrt{\frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}} \quad (2.11)$$

4. Decision rule:

$$H_1 : \mu_x \neq \mu_y \rightarrow \text{Reject } H_0 \text{ if: } |t| > t_{1-\alpha/2, df}$$

$$H_2 : \mu_x > \mu_y \rightarrow \text{Reject } H_0 \text{ if: } t > t_{1-\alpha, df}$$

$$H_3 : \mu_x < \mu_y \rightarrow \text{Reject } H_0 \text{ if: } t < -t_{1-\alpha, df}$$

where $t_{1-\alpha/2, n-1}$ is the $1-\alpha/2$ quantile of a Student's t distribution with df degrees of freedom. A lookup table of the Student's t distribution can be found in Table A.2 in the Appendix.

If the variance of the two samples is different, follow the steps outlined above with the adapted equations (Welch's correction) for df and t :

$$df = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m} \right)^2}{\frac{(s_x^2/n)^2}{n-1} + \frac{(s_y^2/m)^2}{m-1}} \quad (2.12)$$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}} \quad (2.13)$$

2.4.3 Two-Sample Permutation Test

The permutation test of differences in means is applicable to skewed and moderately sized datasets and avoids the assumption about the distribution as required for the t -test. The test computes either all the possible test results that could be computed from the observed data or a large random selection of those results. Then it computes which proportion of the results is equal to or more extreme than the result obtained with the samples tested.

The null hypothesis H_0 assumes that group assignment is arbitrary and that there is no difference in the means, since all observations come from the same population.

Computational steps:

1. Rearrange the data regardless of group assignment in either all possible rearrangements or in several randomly selected ones.
2. Compute the difference observed in the two group means $\bar{x} - \bar{y}$ (test statistic) for each rearrangement. This represents the distribution of the distribution of differences in mean if the null hypothesis H_0 were true.
3. The p -value is the proportion of rearrangements that is outside (or below / above for 1-sided test) the observed difference. The ranking of the observed test statistic among the shuffled values gives the p -value.

2.4.4 Test for the Equality of Variance - Parametric

To check if two independent data samples have the same variance, as for example assumed for the two-sample t test without the Welch's correction, the following test can be employed. The null hypothesis assumes that both samples x and y have the same variance. The test assumes a normal distribution of the observations in both groups.

$$\begin{aligned} H_0 : & \sigma_x^2 = \sigma_y^2 \\ H_1 : & \sigma_x^2 \neq \sigma_y^2 \end{aligned}$$

Computational Steps:

1. For both groups, compute the variance s_x^2 and s_y^2 , where s_x^2 is the greater of the two values.
2. Compute the test statistic T :

$$T = \frac{s_x^2}{s_y^2} \quad (2.14)$$

3. Decision rule:

$$\text{Reject } H_0 \text{ if: } T > F_{1-\alpha, n-1, m-1}$$

where $F_{1-\alpha, n-1, m-1}$ is the $1 - \alpha$ quantile of the F distribution with $n - 1$ (n is the sample size of the sample with the larger variance) and $m - 1$ (m is the sample size of the sample with the smaller variance) degrees of freedom. A lookup table of the F distribution can be found in Table A.4 in the Appendix.

2.5 Comparison among Several Data Groups

These tests allow comparison among more than two groups of data. These tests are applicable with continuous data, such as concentrations or water levels, when the distribution of the data in one group may differ from one or more of the other groups.

2.5.1 Kruskal-Wallis Test - Nonparametric

As for other nonparametric tests, the test results are not affected by a monotonic transformation of the data. No assumption about the distribution is made. The test can be used instead of the ANOVA test when the data is not normally distributed or the assumption of equal variance is not valid.

$$\begin{aligned} H_0 : & \text{All } k \text{ groups have identical distributions.} \\ H_1 : & \text{At least one group has a different distribution.} \end{aligned}$$

Computational steps:

1. Assign ranks to the data (use average ranking for ties), where 1 is the rank for the lowest and N the rank for the highest value.

$$N = \sum_{j=1}^k n_j \quad (2.15)$$

where n_j is the sample size of each of the k groups and N represents the total number of observations.

2. For each of the k groups, compute the average rank $\bar{R}_j$ of the n_j data values in that group.
3. Compute the test statistic KW:

$$KW = \frac{12}{N(N+1)} \sum_{j=1}^k n_j \left(\bar{R}_j - \frac{N+1}{2} \right)^2 \quad (2.16)$$

4. Decision rule:

$$\text{Reject } H_0 \text{ if: } KW \geq \chi^2_{1-\alpha, k-1}$$

where $\chi^2_{1-\alpha, k-1}$ represents the $1 - \alpha$ quantile of the chi-square distribution with $k - 1$ degrees of freedom. A lookup table of the χ^2 distribution can be found in Table A.3 in the Appendix.

An exact form exists for the comparison of three groups with samples size ≤ 5 , or for four or more groups of size four or less per group.

2.5.2 Analysis of Variance ANOVA - Parametric

Analysis of variance ANOVA determines whether all groups have identical mean values by comparing two estimates of the overall variance. If the null hypothesis is true, group means will differ only slightly from the overall mean. The total variance in the data will therefore be very similar to the variance within a group around that group mean. If group mean are dissimilar, some of them will be sufficiently different from the overall mean that the variance within groups no longer equals the total variance.

H_0 : All k groups have identical means.

H_1 : The mean of at least one group is different.

For the test to be applied, all sample groups are independent of each other and each groups consists of a random sample from its population. The data have to obey normality and have the same variance for all groups. The violation of these assumptions results in a loss of the ability to identify differences between the means, which is a loss in power.

The within-group variance is estimated by the mean square error MSE. The treatment mean square MST estimates the sum of the within group and among group variances. The treatment and error mean squares are computed as their sum of squares divided by their degree of freedom df .

Computational steps:

1. Compute the MST and MSE as

$$MST = \frac{\sum_{j=1}^k n_j (\bar{y}_j - \bar{y})^2}{k - 1} = \frac{SST}{k - 1} \quad (2.17)$$

$$MSE = \frac{\sum_{j=1}^k \sum_{i=1}^{n_j} (y_{i,j} - \bar{y}_j)^2}{N - k} = \frac{SSE}{N - k} \quad (2.18)$$

where $\bar{y}_j$ is the j th group mean, n_j is the number of observations in group j , and $\bar{y}$ is the overall mean.

2. Compute the test statistic $f = MST/MSE$.

3. Decision rule:

$$\text{Reject } H_0 \text{ if: } f \geq F_{1-\alpha, k-1, N-k}$$

where $F_{1-\alpha, k-1, N-k}$ is the $1 - \alpha$ quantile of the F distribution with $k - 1$ and $N - k$ degrees of freedom. A lookup table of the F distribution can be found in Table A.4 in the Appendix.

Welch's correction for unequal variance in t tests was also extended to ANOVA. If it is available in software, use it in all cases. In Welch's adjusted ANOVA, the f statistic is computed by weighing each group's contribution to MST by n/s^2 , which means smaller weights for groups with higher variance. An adjusted number of degrees of freedom is used in the MSE.

2.5.3 One-way Permutation Test

A permutation test for the difference in means is an alternative to the ANOVA test and should be implemented when the data is skewed. It operates on the null hypothesis that a all observations even in different groups come from the same population.

H_0 : All k groups have identical means.

H_0 : The mean of at least one group is different.

To perform the test, randomize the observations into groups (same number of observations as the original groups). For each randomization, compute the ANOVA test statistic. Construct the test statistic distribution by repeating the randomization for all possible combinations (for small sample sizes) or thousands of random combinations representing a large selection of possible rearrangements.

The p-value is the proportion that equals or exceeds the observed value of f .

For tests on three or more dependent groups or two-factor ANOVA tests, see Chapter 7 in the USGS Handbook on the Statistical Methods in Water Resources (Helsel and Hirsch 2002).

3 Goodness-of-Fit

Goodness-of-fit tests play a central role in stochastic hydrology by helping determine whether a proposed probability distribution adequately represents observed hydrological data. Because many hydrological analyses — such as flood frequency estimation, rainfall modeling, and drought assessment — rely on assumed statistical distributions, it is essential to evaluate how well these assumptions match empirical observations. Goodness-of-fit procedures provide formal statistical tools for comparing theoretical models with measured data and identifying significant discrepancies. This chapter introduces the principles of goodness-of-fit testing and presents commonly used tests applied in hydrological practice.

Analytical goodness-of-fit criteria, such as the Chi-square test and the Kolmogorov–Smirnov test, quantify the agreement between observed data and a theoretical distribution through formal statistical measures and decision rules. In contrast, probability plots provide a graphical approach, allowing the analyst to visually assess deviations from the assumed distribution and to identify systematic patterns that may not be fully captured by numerical test statistics.

3.1 Goodness-of-Fit Tests

3.1.1 χ^2 Test

The χ^2 test assesses how close the observations are to the values expected under the fitted distribution model.

H_0 : The pdf of the tested distribution fits the data.

H_1 : The pdf of the tested distribution does not fit the data.

Computational steps:

1. Divide the observations in classes of equal length or equal probability.
2. Compute the observed O_i and expected E_i number of observations in each class i . E_i is computed from the fitted theoretical distribution.
3. Compute the test statistic χ_c^2 :

$$\chi_c^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (3.1)$$

where k represents the number of classes.

4. Decision rule:

$$\text{Reject } H_0 \text{ if: } \chi_c^2 > \chi_{1-\alpha, \nu}^2$$

where $\chi_{1-\alpha, \nu}^2$ represents the $1 - \alpha$ quantile of the χ^2 distribution with ν degrees of freedom. ν is given by $\nu = k - 1 - p$, where p represents the number of estimated parameters.

3.1.2 Kolmogoro-Smirnov Test

The Kolmogorov-Smirnov Test KS operates on the same null and alternative hypothesis as the χ^2 test, but uses the empirical and fitted cumulative distribution function instead of the histogram and the probability density function.

Computational steps:

1. Compute the empirical cumulative distribution function $F_n(x) = i/n$ for all observations x_i , where i represents the rank in ascending order and n represents the total number of observations.
2. Compute the theoretical cumulative distribution function $F_X(x)$ by evaluating it at each observed value x_i .
3. Compute the test statistic D_n :

$$D_n = \max |F_n(x) - F_X(x)| \quad (3.2)$$

4. Decision rule:

$$\text{Reject } H_0 \text{ if: } D_n \geq C_\alpha$$

The critical value of the Kolmogorov-Smirnov test C_α is computed as

$$C_\alpha = \frac{K_\alpha}{\sqrt{n} + 0.12 + 0.11/\sqrt{n}} \quad (3.3)$$

where the value of K_α depends on the significance level α : $K_\alpha(\alpha = 0.10) = 1.224$, $K_\alpha(\alpha = 0.05) = 1.358$, and $K_\alpha(\alpha = 0.01) = 1.628$.

For small samples with $n < 30$, if the distribution parameters have been fit with the same data used for the test, a correction is needed (see Lillefors test).

3.1.3 Probability Plot Corelation Coefficient Test PPCC

The probability plot correlation coefficient tests the correlation between the ordered observations and their fitted quantiles under the chosen distribution.

Computational steps:

1. Sort the observations x_i in ascending order $i = 1, \dots, n$.
2. For each rank i compute the associated plotting position q_i (select the adequate equation from Tab. 3.1). This represents the empirical non-exceedance probability $F_n(x_i)$.
3. For each observation, compute the fitted quantile w_i , which represents the theoretical observation for the given plotting position as

$$r = \frac{\sum_i (x_i - \bar{x})(w_i - \bar{w})}{\sqrt{\sum_i (x_i - \bar{x})^2 \cdot \sum_i (w_i - \bar{w})^2}} \quad (3.4)$$

where $\bar{x}$ represents the average value of the observations and $\bar{w}$ represents the average value of the fitted quantiles.

4. Decision rule:

$$\text{Reject } H_0 \text{ if: } r < r_{n, \alpha}$$

where $r_{n, \alpha}$ represents the α quantile for n observations of the PPCC test distribution. Depending on the distribution tested for, these values differ.

3.2 Probability Plots

Probability plots represent a graphical goodness of fit test. Given x_i observations with $i = 1, \dots, N$, we plot the ranked observations (e.g. smallest to largest) with their expected value $F_X^{-1}(q_i)$, where q_i is the plotting position (cumulative probability) under the tested probability distribution. A selection of different equations for the calculation of the plotting position q_i is presented in Table 3.1.

Table 3.1. Summary of different plotting positions, including their equations for q_i and comments on the intended use.

Plotting position	Equation	Comment
Weibull	$\frac{i}{N+1}$	Unbiased exceedance probability for all distributions
Gringorton	$\frac{i - 0.44}{N + 0.12}$	Optimized for the Gumbel distribution
Blom	$\frac{i - 3/8}{N + 1/4}$	Optimized for the Normal distribution

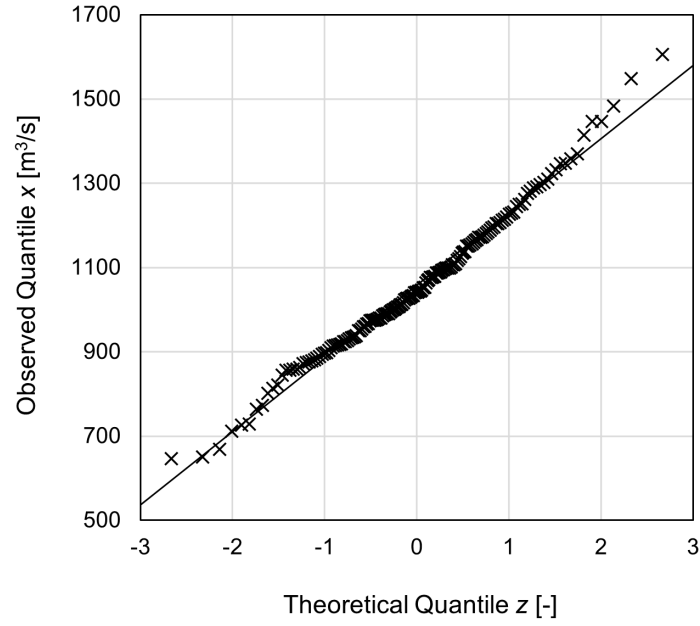


Figure 3.1. Quantile-quantile plot of annual precipitation sums at Zürich Fluntern measuring station from 1865-2024 testing for its normal distribution. The theoretical quantile on the x-axis is represented by the standard normal variate $z_i = \Phi^{-1}(q_i)$, where the plotting position q_i is computed using Blom's equation (see Tab. 3.1). On the y-axis, the observed quantile is represented by the actual observations x_i . The 1:1 line is therefore described by $x = \sigma_X \cdot z + \bar{x}$, where $\bar{x}$ and σ_X represent the fitted parameters of the normal distribution.

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A Appendix

A.1 Tables of Critical Values

Table A.1. Tabled values for the normal standard distribution, relating the curve left of the reduced variate z under the curve. For values with $z < 0$, employ the relation $\Phi(-z) = 1 - \Phi(z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997

Table A.2. Table of the Student's t distribution for different degrees of freedom df and significance levels α .

df	α level									
	0.25	0.2	0.15	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
5	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
6	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
7	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
8	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
9	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
10	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
11	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
12	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
13	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
14	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
15	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
16	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
17	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
18	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
19	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
20	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
21	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
22	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
23	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
24	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
25	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
26	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
27	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
28	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
29	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
30	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
40	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
60	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
80	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
100	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
1000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table A.3. Table of the χ^2 distribution for different degrees of freedom df and significance levels α .

df	α level									
	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Table A.4. Table of the $F_{1-\alpha, n, m}$ distribution for different values of the degrees of freedom in the numerator n and the denominator m , and the significance levels α .

$1 - \alpha$	n	1	2	3	4	5	6	m 8	10	15	20	30	60	120
0.9	1	39.9	49.5	53.6	55.8	57.2	58.2	59.4	60.2	61.2	61.7	62.3	62.8	63.1
0.95		161	200	216	225	230	234	239	242	246	248	250	252	253
0.975		648	799	864	900	922	937	957	969	985	993	1001	1010	1014
0.99		4052	5000	5403	5625	5764	5859	5981	6056	6157	6209	6261	6313	6339
0.995		16211	20000	21615	22500	23056	23437	23925	24224	24630	24836	25044	25253	25359
0.9	2	8.526	9.000	9.162	9.243	9.293	9.326	9.367	9.392	9.425	9.441	9.458	9.475	9.483
0.95		18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.40	19.43	19.45	19.46	19.48	19.49
0.975		38.51	39.00	39.17	39.25	39.30	39.33	39.37	39.40	39.43	39.45	39.46	39.48	39.49
0.99		98.50	99.00	99.17	99.25	99.30	99.33	99.37	99.40	99.43	99.45	99.47	99.48	99.49
0.995		198.50	199.00	199.17	199.25	199.30	199.33	199.37	199.40	199.43	199.45	199.47	199.48	199.49
0.9	3	5.538	5.462	5.391	5.343	5.309	5.285	5.252	5.230	5.200	5.184	5.168	5.151	5.143
0.95		10.128	9.552	9.277	9.117	9.013	8.941	8.845	8.786	8.703	8.660	8.617	8.572	8.549
0.975		17.44	16.04	15.44	15.10	14.88	14.73	14.54	14.42	14.25	14.17	14.08	13.99	13.95
0.99		34.12	30.82	29.46	28.71	28.24	27.91	27.49	27.23	26.87	26.69	26.50	26.32	26.22
0.995		55.55	49.80	47.47	46.19	45.39	44.84	44.13	43.69	43.08	42.78	42.47	42.15	41.99
0.9	4	4.545	4.325	4.191	4.107	4.051	4.010	3.955	3.920	3.870	3.844	3.817	3.790	3.775
0.95		7.709	6.944	6.591	6.388	6.256	6.163	6.041	5.964	5.858	5.803	5.746	5.688	5.658
0.975		12.218	10.649	9.979	9.605	9.364	9.197	8.980	8.844	8.657	8.560	8.461	8.360	8.309
0.99		21.20	18.00	16.69	15.98	15.52	15.21	14.80	14.55	14.20	14.02	13.84	13.65	13.56
0.995		31.33	26.28	24.26	23.15	22.46	21.97	21.35	20.97	20.44	20.17	19.89	19.61	19.47
0.9	5	4.060	3.780	3.619	3.520	3.453	3.405	3.339	3.297	3.238	3.207	3.174	3.140	3.123
0.95		6.608	5.786	5.409	5.192	5.050	4.950	4.818	4.735	4.619	4.558	4.496	4.431	4.398
0.975		10.007	8.434	7.764	7.388	7.146	6.978	6.757	6.619	6.428	6.329	6.227	6.123	6.069
0.99		16.258	13.274	12.060	11.392	10.967	10.672	10.289	10.051	9.722	9.553	9.379	9.202	9.112
0.995		22.78	18.31	16.53	15.56	14.94	14.51	13.96	13.62	13.15	12.90	12.66	12.40	12.27
0.9	6	3.776	3.463	3.289	3.181	3.108	3.055	2.983	2.937	2.871	2.836	2.800	2.762	2.742
0.95		5.987	5.143	4.757	4.534	4.387	4.284	4.147	4.060	3.938	3.874	3.808	3.740	3.705
0.975		8.813	7.260	6.599	6.227	5.988	5.820	5.600	5.461	5.269	5.168	5.065	4.959	4.904
0.99		13.745	10.925	9.780	9.148	8.746	8.466	8.102	7.874	7.559	7.396	7.229	7.057	6.969
0.995		18.635	14.544	12.917	12.028	11.464	11.073	10.566	10.250	9.814	9.589	9.358	9.122	9.001
0.9	8	3.458	3.113	2.924	2.806	2.726	2.668	2.589	2.538	2.464	2.425	2.383	2.339	2.316
0.95		5.318	4.459	4.066	3.838	3.687	3.581	3.438	3.347	3.218	3.150	3.079	3.005	2.967
0.975		7.571	6.059	5.416	5.053	4.817	4.652	4.433	4.295	4.101	3.999	3.894	3.784	3.728
0.99		11.259	8.649	7.591	7.006	6.632	6.371	6.029	5.814	5.515	5.359	5.198	5.032	4.946
0.995		14.688	11.042	9.596	8.805	8.302	7.952	7.496	7.211	6.814	6.608	6.396	6.177	6.065
0.9	10	3.285	2.924	2.728	2.605	2.522	2.461	2.377	2.323	2.244	2.201	2.155	2.107	2.082
0.95		4.965	4.103	3.708	3.478	3.326	3.217	3.072	2.978	2.845	2.774	2.700	2.621	2.580
0.975		6.937	5.456	4.826	4.468	4.236	4.072	3.855	3.717	3.522	3.419	3.311	3.198	3.140
0.99		10.044	7.559	6.552	5.994	5.636	5.386	5.057	4.849	4.558	4.405	4.247	4.082	3.996
0.995		12.826	9.427	8.081	7.343	6.872	6.545	6.116	5.847	5.471	5.274	5.071	4.859	4.750
0.9	15	3.073	2.695	2.490	2.361	2.273	2.208	2.119	2.059	1.972	1.924	1.873	1.817	1.787
0.95		4.543	3.682	3.287	3.056	2.901	2.790	2.641	2.544	2.403	2.328	2.247	2.160	2.114
0.975		6.200	4.765	4.153	3.804	3.576	3.415	3.199	3.060	2.862	2.756	2.644	2.524	2.461
0.99		8.683	6.359	5.417	4.893	4.556	4.318	4.004	3.805	3.522	3.372	3.214	3.047	2.959
0.995		10.798	7.701	6.476	5.803	5.372	5.071	4.674	4.424	4.070	3.883	3.687	3.480	3.372
0.9	20	2.975	2.589	2.380	2.249	2.158	2.091	1.999	1.937	1.845	1.794	1.738	1.677	1.643
0.95		4.351	3.493	3.098	2.866	2.711	2.599	2.447	2.348	2.203	2.124	2.039	1.946	1.896
0.975		5.871	4.461	3.859	3.515	3.289	3.128	2.913	2.774	2.573	2.464	2.349	2.223	2.156
0.99		8.096	5.849	4.938	4.431	4.103	3.871	3.564	3.368	3.088	2.938	2.778	2.608	2.517
0.995		9.944	6.986	5.818	5.174	4.762	4.472	4.090	3.847	3.502	3.318	3.123	2.916	2.806
0.9	30	2.881	2.489	2.276	2.142	2.049	1.980	1.884	1.819	1.722	1.667	1.606	1.538	1.499
0.95		4.171	3.316	2.922	2.690	2.534	2.421	2.266	2.165	2.015	1.932	1.841	1.740	1.683
0.975		5.568	4.182	3.589	3.250	3.026	2.867	2.651	2.511	2.307	2.195	2.074	1.940	1.866
0.99		7.562	5.390	4.510	4.018	3.699	3.473	3.173	2.979	2.700	2.549	2.386	2.208	2.111
0.995		9.180	6.355	5.239	4.623	4.228	3.949	3.580	3.344	3.006	2.823	2.628	2.415	2.300
0.9	60	2.791	2.393	2.177	2.041	1.946	1.875	1.775	1.707	1.603	1.543	1.476	1.395	1.348
0.95		4.001	3.150	2.758	2.525	2.368	2.254	2.097	1.993	1.836	1.748	1.649	1.534	1.467
0.975		5.286	3.925	3.343	3.008	2.786	2.627	2.412	2.270	2.061	1.944	1.815	1.667	1.581
0.99		7.077	4.977	4.126	3.649	3.339	3.119	2.823	2.632	2.352	2.198	2.028	1.836	1.726
0.995		8.495	5.795	4.729	4.140	3.760	3.492	3.134	2.904	2.570	2.387	2.187	1.962	1.834
0.9	120	2.748	2.347	2.130	1.992	1.896	1.824	1.722	1.652	1.545	1.482	1.409	1.320	1.265
0.95		3.920	3.072	2.680	2.447	2.290	2.175	2.016	1.910	1.750	1.659	1.554	1.429	1.352
0.975		5.152	3.805	3.227	2.894	2.674	2.515	2.299	2.157	1.945	1.825	1.690	1.530	1.433
0.99		6.851	4.787	3.949	3.480	3.174	2.956	2.663	2.472	2.192	2.035	1.860	1.656	1.533
0.995		8.179	5.539	4.497	3.921	3.548	3.285	2.933	2.705	2.373	2.188	1.984	1.747	1.606

Figure A.1. Exact values of the signed-rank critical test statistic x and x' . For numbers of observation $n > 15$ and $\alpha < 0.025$, the large sample approximation in Chapter 2.3.1 can be used. The left table displays values of $x'(\alpha, n)$ representing the low tail of the distribution, the table on the right displays $x(\alpha, n)$ representing the upper tail of the distribution. (Helsel and Hirsch 2002, Table B6)

n	α -level				n	α -level			
	.005	.010	.025	.050		.005	.010	.025	.050
5				0	5				15
6			0	2	6			21	19
7		0	2	3	7		28	26	25
8	0	1	3	5	8	36	35	33	31
9	1	3	5	8	9	44	42	40	37
10	3	5	8	10	10	52	50	47	45
11	5	7	10	13	11	61	59	56	53
12	7	9	13	17	12	71	69	65	61
13	9	12	17	21	13	82	79	74	70
14	12	15	21	25	14	93	90	84	80
15	15	19	25	30	15	105	101	95	90
16	19	23	29	35	16	117	113	107	101
17	23	27	34	41	17	130	126	119	112
18	27	32	40	47	18	144	139	131	124
19	32	37	46	53	19	158	153	144	137
20	37	43	52	60	20	173	167	158	150

Figure A.2. Exact values of the rank-sum critical test statistics x and x^* for different sample sizes m and n , and probabilities. When finding the critical value $x = x(\alpha, n, m)$, the column marked as p represents the significance level α . (Helsel and Hirsch 2002, Table B4)

n [smaller sample size] = 3									
m=4		m=5		m=6		m=7		m=8	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
16.114	8	18.125	9	20.131	10	22.133	11	24.139	12
17.057	7	19.071	8	21.083	9	23.092	10	25.097	11
18.029	6	20.036	7	22.048	8	24.058	9	26.067	10
19.0	5	21.018	6	23.024	7	25.033	8	27.042	9
		22.0	5	24.012	6	26.017	7	28.024	8
				25.0	5	27.008	6	29.012	7
						28.0	5	30.006	6
								33.005	6
n [smaller sample size] = 4									
m=4		m=5		m=6		m=7		m=8	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
22.171	14	25.143	15	28.129	16	31.115	17	34.107	18
23.100	13	26.095	14	29.086	15	32.082	16	35.077	17
24.057	12	27.056	13	30.057	14	33.055	15	36.055	16
25.029	11	28.032	12	31.033	13	34.036	14	37.036	15
26.014	10	29.016	11	32.019	12	35.021	13	38.024	14
27.0	9	30.008	10	33.010	11	36.012	12	39.014	13
		31.0	9	34.005	10	37.006	11	40.008	12
						38.003	10	41.004	11
								44.006	12
n [smaller sample size] = 5									
m=5		m=6		m=7		m=8		m=9	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
34.111	21	37.123	23	41.101	24	44.111	26	47.120	28
35.075	20	38.089	22	42.074	23	45.085	25	48.095	27
36.048	19	39.063	21	43.053	22	46.064	24	49.073	26
37.028	18	40.041	20	44.037	21	47.047	23	50.056	25
38.016	17	41.026	19	45.024	20	48.033	22	51.041	24
39.008	16	42.015	18	46.015	19	49.023	21	52.030	23
40.004	15	43.009	17	47.009	18	50.015	20	53.021	22
		44.004	18	48.005	17	51.009	19	54.014	21
						52.005	18	55.009	20
								56.006	19
n [smaller sample size] = 6									
m=6		m=7		m=8		m=9		m=10	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
47.120	31	51.117	33	55.114	35	59.112	37	63.110	39
48.090	30	52.090	32	56.091	34	60.091	36	64.090	38
49.066	29	53.069	31	57.071	33	61.072	35	65.074	37
50.047	28	54.051	30	58.054	32	62.057	34	66.059	36
51.032	27	55.037	29	59.041	31	63.044	33	67.047	35
52.021	26	56.026	28	60.030	30	64.033	32	68.036	34
53.013	25	57.017	27	61.021	29	65.025	31	69.028	33
54.008	24	58.011	26	62.015	28	66.018	30	70.021	32
		59.007	25	63.010	27	67.013	29	71.016	31
						68.009	28	72.011	30
								73.008	29
n [smaller sample size] = 7									
m=7		m=8		m=9		m=10		m=11	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
61.159	44	65.168	47	70.150	49	74.157	52	78.164	55
62.130	43	66.140	46	71.126	48	75.135	51	79.141	54
63.104	42	67.116	45	72.105	47	76.115	50	80.118	53
64.082	41	68.095	44	73.087	46	77.097	49	81.097	52
65.064	40	69.076	43	74.071	45	78.081	48	82.079	51
66.049	39	70.060	42	75.057	44	79.067	47	83.063	50
67.036	38	71.047	41	76.045	43	80.054	46	84.049	49
68.027	37	72.036	40	77.036	42	81.044	45	85.037	48
69.019	36	73.027	39	78.027	41	82.035	44	86.028	47
70.013	35	74.020	38	79.021	40	83.028	43	87.021	46
71.009	34	75.014	37	80.016	39	84.024	42	88.016	45
72.006	33	76.010	36	81.011	38	85.021	41	89.012	44
		77.007	35	82.008	37	86.012	40	90.009	43
				83.006	36	87.009	39		
n [smaller sample size] = 8									
m=8		m=9		m=10		m=11		m=12	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
79.13957		84.138	60	89.137	63	94.137	66	99.137	69
80.11756		85.118	59	90.118	62	95.118	65	100.118	68
81.09755		86.100	58	91.102	61	96.102	64	101.102	67
82.08054		87.084	57	92.086	60	97.086	63	102.086	66
83.06553		88.069	56	93.073	59	98.073	62	103.073	65
84.05252		89.057	55	94.061	58	99.061	61	104.061	64
85.04151		90.046	54	95.051	57	100.051	60	105.051	63
86.03250		91.037	53	96.042	56	101.042	59	106.042	62
87.02549		92.030	52	97.034	55	102.034	58	107.034	61
88.01948		93.023	51	98.027	54	103.027	57	108.027	60
89.01447		94.018	50	99.022	53	104.022	56	109.022	59
90.01046		95.014	49	100.017	52	105.017	55	110.017	58
91.00745		96.010	48	101.013	51	106.013	54	111.013	57
92.00544		97.008	47	102.010	50	107.010	53	112.010	56
n [smaller sample size] = 10									
m=10		m=11		m=12		m=13		m=14	
x	x^*	x	x^*	x	x^*	x	x^*	x	x^*
119.157	91	120.140	90	121.124	89	122.109	88	123.095	87
120.140	90	121.124	89	122.109	88	123.095	87	124.083	86
121.124	89	122.109	88	123.095	87	124.083	86	125.072	85
122.109	88	123.095	87	124.083	86	125.072	85	126.062	84
123.095	87	124.083	86	125.072	85	126.062	84	127.053	83
124.083	86	125.072	85	126.062	84	127.053	83	128.045	82
125.072	85	126.062	84	127.053	83	128.045	82	129.038	81
126.062	84	127.053	83	128.045	82	129.038	81	130.032	80
127.053	83	128.045	82	129.038	81	130.032	80	131.026	79
128.045	82	129.038	81	130.032	80	131.026	79	132.022	78
129.038	81	130.032	80	131.026	79	132.022	78	133.018	77
130.032	80	131.026	79	132.022	78	133.018	77	134.014	76
131.026	79	132.022	78	133.018	77	134.014	76	135.012	75
132.022	78	133.018	77	134.014	76	135.012	75	136.009	74
133.018	77	134.014	76	135.012	75	136.009	74	137.007	73
134.014	76	135.012	75	136.009	74	137.007	73	138.006	72
135.012	75	136.009	74	137.007	73	138.006	72		
136.009	74	137.007	73	138.006	72				
137.007	73								
138.006	72								

Table A.5. Something

n	alpha level						n	alpha level					
	0.005	0.01	0.025	0.05	0.1	0.25		0.005	0.01	0.025	0.05	0.1	0.25
3	0.867	0.869	0.872	0.879	0.891	0.924	32	0.943	0.950	0.959	0.966	0.972	0.980
4	0.813	0.824	0.846	0.868	0.894	0.931	33	0.944	0.951	0.961	0.967	0.973	0.981
5	0.807	0.826	0.856	0.880	0.903	0.934	34	0.946	0.953	0.962	0.968	0.974	0.981
6	0.820	0.838	0.866	0.888	0.910	0.939	35	0.947	0.954	0.962	0.969	0.974	0.982
7	0.828	0.850	0.877	0.898	0.918	0.944	36	0.948	0.955	0.963	0.969	0.975	0.982
8	0.840	0.861	0.887	0.906	0.924	0.948	37	0.950	0.956	0.964	0.970	0.976	0.983
9	0.854	0.871	0.894	0.912	0.930	0.952	38	0.951	0.957	0.965	0.971	0.976	0.983
10	0.862	0.879	0.901	0.918	0.934	0.954	39	0.951	0.958	0.966	0.971	0.977	0.983
11	0.870	0.886	0.907	0.923	0.938	0.957	40	0.953	0.959	0.966	0.972	0.977	0.984
12	0.876	0.892	0.912	0.928	0.942	0.960	41	0.953	0.960	0.967	0.973	0.977	0.984
13	0.885	0.899	0.918	0.932	0.945	0.962	42	0.954	0.961	0.968	0.973	0.978	0.984
14	0.890	0.905	0.923	0.935	0.948	0.964	43	0.956	0.961	0.968	0.974	0.978	0.984
15	0.896	0.910	0.927	0.939	0.951	0.965	44	0.957	0.962	0.969	0.974	0.979	0.985
16	0.899	0.913	0.929	0.941	0.953	0.967	45	0.957	0.963	0.969	0.974	0.979	0.985
17	0.905	0.917	0.932	0.944	0.954	0.968	46	0.958	0.963	0.970	0.975	0.980	0.985
18	0.908	0.920	0.935	0.946	0.957	0.970	47	0.959	0.965	0.971	0.976	0.980	0.986
19	0.914	0.924	0.938	0.949	0.958	0.971	48	0.959	0.965	0.971	0.976	0.980	0.986
20	0.916	0.926	0.940	0.951	0.960	0.972	49	0.961	0.966	0.972	0.976	0.981	0.986
21	0.918	0.930	0.943	0.952	0.961	0.973	50	0.961	0.966	0.972	0.977	0.981	0.986
22	0.923	0.933	0.945	0.954	0.963	0.974	55	0.965	0.969	0.974	0.979	0.982	0.987
23	0.925	0.935	0.947	0.956	0.964	0.975	60	0.967	0.971	0.976	0.980	0.984	0.988
24	0.927	0.937	0.949	0.957	0.965	0.976	65	0.969	0.973	0.978	0.981	0.985	0.989
25	0.929	0.939	0.951	0.959	0.966	0.976	70	0.971	0.975	0.979	0.983	0.986	0.990
26	0.932	0.941	0.952	0.960	0.967	0.977	75	0.973	0.976	0.981	0.984	0.987	0.990
27	0.934	0.943	0.953	0.961	0.968	0.978	80	0.975	0.978	0.982	0.985	0.987	0.991
28	0.936	0.944	0.955	0.962	0.969	0.978	85	0.976	0.979	0.983	0.985	0.988	0.991
29	0.939	0.946	0.956	0.963	0.970	0.979	90	0.977	0.980	0.984	0.986	0.988	0.992
30	0.939	0.947	0.957	0.964	0.971	0.979	95	0.979	0.981	0.984	0.987	0.989	0.992
31	0.942	0.950	0.958	0.965	0.972	0.980	100	0.979	0.982	0.985	0.987	0.989	0.992